BIO-RAD

Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Species Used *Staphylococcus aureus*

Molecules Electroporated DNA: plasmids, pC194, pE194, pTV32; phage DNA, 80α.

Before the Pulse

Cell Growth Medium Tryptic Soy Broth (TSB - Difco)

Growth Phase at Harvest O.D. (540) = 0.55 to 0.6

Pre-pulse Incubation 30 min. ice with DNA

Wash Solution 10% glycerol

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 0 °C

Electroporation Medium Not given

Cuvette Gap 0.2 cm

Cell Density 1 X 10 (11) cells / ml

Voltage 2.5 kV

Volume of Cells 50 µl

Field Strength 12.5 kV/cm

DNA Concentration 0.1 to 0.2 µg / ml

Capacitor 25 µF

DNA Resuspension Buffer 5 parts 4X SOC, 4 parts SMM, 0.5 parts 10% BSA + DNA

Resistor (Pulse Controller)100 Ω

Volume of DNA 10% volume, ≤ 5 µl

Time Constant 2.4 msec

After the Pulse

Outgrowth Medium 4X SOC, SMM,10%

Outgrowth Temperature 37 °C (30°C for sensitive plasmids)

Length of Incubation 90 min.

Selection Method or Assay Used BM agar + antibiotics

Electroporation Efficiency 10 (5) transformants / µg

Per Cent Survival 80%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
SMM :1M sucrose, 0.04 M maleic, 0.04 MgCl2.
SMM + 4X SOC: 5.5 parts SMM 4 parts 4X SOC, 0.5 parts 10 % BSA, filter sterilized.
BM: 1.0% peptone, 0.5% yeast extract 0.1 % glucose, 0.5% NaCl, 0.1% K2HPO4 , 1.2 % agar.
4X SOC: 8.0% tryptone, 2.0% yeast extract, 40mM NaCl, 10 mM MgCl2, 40 mM MgSO4, 80mM glucose.
To the freshly autoclaved base containing tyrptone and yeast extract, add the appropriate amount of individually prepared sterile stocks (2M for glucose, 1M for all inorganic salts). Filter sterilize and aspetically tube.

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